

# Lokesh Balasubramanian

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## EDUCATION

New York University, CDS

2026 – 2028

*M.S. in Data Science*

Indian Institute of Technology Delhi

2022 – 2026

*B.Tech. in Engineering Physics*

GPA: 8.3 / 10.0

**Selected Coursework:** Machine Learning, Reinforcement Learning, Independent Study in AI, Linear Algebra, Probability & Statistics, Numerical & Computational Methods, Signals & Systems, Computational Physics, Statistical Physics

## RESEARCH EXPERIENCE

**Yardi School of Artificial Intelligence, IIT Delhi**

*Undergraduate Researcher (Advisor: Prof. Sumeet Agarwal)*

- Investigated the statistical physics of reasoning in large language models by modelling hidden-state trajectories as dynamical systems using DeepSeek-R1-Distill-Llama-70B and DeepSeek-R1-Distill-Qwen-7B.
- Developed an end-to-end pipeline for extracting sentence-level hidden representations and modelling reasoning dynamics using PCA, ridge regression, Gaussian mixture models, and Switching Linear Dynamical Systems (SLDS).
- Evaluated reasoning behaviour on GSM8K and OpenBookQA, conducting large-scale experiments on IIT Delhi's HPC cluster to analyse latent reasoning regimes and their relationship with model performance.

**University of Oxford & CNRS-IAP, Paris**

*Research Intern (Advisor: Dr. Deaglan Bartlett)*

- Conducted research on symbolic regression for cosmological Halo Mass Functions (HMFs), aiming to derive compact, interpretable analytical expressions directly from large-scale cosmological simulations.
- Developed scalable symbolic optimisation pipelines using PyOperon, exploring physically meaningful functional forms and benchmarking against state-of-the-art HMF fitting models.
- Ongoing collaboration on the *SYREN-HMF* project, with work being prepared for publication.

**Department of Physics & CSE, IIT Delhi**

*Undergraduate Researcher (Advisors: Prof. Abhishek Iyer and Prof. Srikanta Bedathur)*

- Conducted research on machine learning for jet tagging in simulated LHC collisions, addressing both single-jet (tau identification) and dijet signal-background classification tasks.
- Generated Monte Carlo simulation datasets and developed end-to-end training pipelines on IIT Delhi's HPC cluster, implementing ParticleNet, Particle Transformer, and FROCC architectures in PyTorch.
- Benchmarked graph neural network, transformer, and Lorentz-equivariant models across jet-tagging tasks, evaluating classification performance on large-scale simulated collider datasets.

## INDUSTRY EXPERIENCE

**Oracle Cloud Infrastructure (OCI), Bangalore**

May 2025 – Jul 2025

*Member of Technical Staff Intern*

- Re-architected a production-critical Reverse Connection Endpoint (RCE) recreation workflow into a parallel, fault-tolerant system, reducing execution time by  $\sim 47\%$ .
- Designed resilient recovery mechanisms including HA-host retries, failure isolation, adaptive health-check filtering, and comprehensive execution reporting, eliminating single-point failures and reducing manual on-call intervention.
- Built a production-ready CLI with flexible batch execution and targeted RCE recreation; validated through unit/integration testing and successfully piloted across multiple OCI regions.

## TECHNICAL SKILLS

**Languages:** Python, C++, Bash | **Tools:** Git, Linux, HPC (PBS/Torque)

**Frameworks & Libraries:** PyTorch, JAX, HF Transformers, HF Datasets, scikit-learn, NumPy, SciPy, pandas, PyOperon

## TEACHING & ACHIEVEMENTS

- **Teaching:** APG Tutor for Quantum Mechanics and Solid State Physics | BSW Student Mentor for incoming undergraduates.
- **Achievements:** JEE Advanced 2022: Top 0.8% nationwide | Qualified for the Regional Mathematical Olympiad (RMO), 2019.